

第十章 绝热近似和 Berry 相位.

1. 绝热定理

绝热定理: 系统哈密顿量缓慢变化,

在初始时刻 t_0 , 系统处在第 n 个本征态 $|n(t_0)\rangle$

则在后续时刻 t , 系统会保持在 $|n(t)\rangle$

绝热近似:

$$\hat{H}(t)|n(t)\rangle = E_n(t)|n(t)\rangle$$

$$|\psi(t)\rangle = \sum_n c_n(t)|n(t)\rangle$$

$$i\hbar \sum_n \left[\left(\frac{\partial}{\partial t} c_n(t) \right) |n(t)\rangle + c_n(t) \frac{\partial}{\partial t} |n(t)\rangle \right] = \sum_n c_n \hat{H}(t) |n(t)\rangle$$

$$\frac{\partial}{\partial t} c_n(t) = \sum_m c_m(t) \left(\hat{H}(t) - i\hbar \frac{\partial}{\partial t} \right) |n(t)\rangle$$

$$\hat{H}(t)|n(t)\rangle = E_n(t)|n(t)\rangle$$

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$$\frac{\partial}{\partial t} c_m(t) = -c_m(t) \langle m(t) | \frac{\partial}{\partial t} | m(t) \rangle - \frac{i}{\hbar} E_m(t) c_m(t)$$

初始时刻 $t=0$

$$c_m(t) = c_m(0) e^{i\Theta_m} e^{i\gamma_m}$$

$$\text{其中 } \Theta_m(t) = -\frac{i}{\hbar} \int_0^t E_m(t') dt' \quad \text{动力学相位}$$

$$\gamma_m(t) = i \int_0^t \langle m(t') | \frac{\partial}{\partial t'} | m(t') \rangle dt' \quad \text{几何相位.}$$

$$\langle m(t) | m(t) \rangle = 1$$

$$\frac{\partial}{\partial t} \langle m(t) | m(t) \rangle = 0$$

$$\langle \frac{\partial}{\partial t} m(t) | m(t) \rangle + \langle m(t) | \frac{\partial}{\partial t} m(t) \rangle = 0$$

$$\uparrow \left(\langle m(t) | \frac{\partial}{\partial t} | m(t) \rangle^* + \langle m(t) | \frac{\partial}{\partial t} | m(t) \rangle \right) = 0$$

§ 10.3 Berry 相位.

设哈密顿量依赖于参数 $\vec{R}(t) = (R_1(t), R_2(t), \dots)$

$$\begin{aligned}\delta_m(t) &= i \int_0^t \langle n(t') | \frac{d}{dt'} | n(t') \rangle dt' \\ &= i \int_{\vec{R}_i}^{\vec{R}_f} \langle n(\vec{R}) | \vec{v}_R | n(\vec{R}) \rangle d\vec{R} \\ &= \int_{\vec{R}_i}^{\vec{R}_f} \vec{A}_n \cdot d\vec{R} \quad \vec{A}_n = i \langle n(\vec{R}) | \vec{v}_R | n(\vec{R}) \rangle\end{aligned}$$

在规范变换 $|n(\vec{R})\rangle \rightarrow |n'(\vec{R})\rangle = e^{i\chi(\vec{R})} |n(\vec{R})\rangle$ 下:

$$\begin{aligned}\vec{A}_n &\rightarrow \vec{A}_n' = i \langle n(\vec{R}) | e^{-i\chi(\vec{R})} \vec{v}_R e^{i\chi(\vec{R})} | n(\vec{R}) \rangle \\ &= \vec{A}_n - \vec{\nabla}_R \chi(\vec{R})\end{aligned}$$

选取参数空间中的闭合回路 C

此时 $\delta_m(C)$ 是规范不变的

$$\delta_m(C) = \oint_C \vec{A}_n \cdot d\vec{R} \quad \text{Berry 联络}$$

$$\begin{aligned}\text{Stokes 定理} &= -2m \iint_C \nabla \times \langle n(\vec{R}) | \vec{v}_R | n(\vec{R}) \rangle \cdot d\vec{S} \\ &= -2m \iint_C (\langle \nabla n | \times | \nabla n \rangle) \cdot d\vec{S} \\ &= -2m \iint_C \sum_m (\langle \nabla n | m \rangle \times \langle m | \nabla n \rangle) \cdot d\vec{S}\end{aligned}$$

$$\text{Berry 曲率} = -2m \iint_C \vec{V}_n \cdot d\vec{S} \quad \vec{V}_n = \sum_m (\langle \nabla n | m \rangle \times \langle m | \nabla n \rangle)$$

$$\hat{H} |n\rangle = \epsilon_n |n\rangle$$

$$\nabla \cdot (\nabla \hat{H}) |n\rangle + \hat{H} | \nabla n \rangle = \nabla \epsilon_n |n\rangle + \epsilon_n | \nabla n \rangle$$

$$\langle m | \nabla n \rangle = \frac{\langle m | \nabla \hat{H} | n \rangle}{\epsilon_n - \epsilon_m}$$

$$\vec{V}_n = 2m \sum_{n \neq m} \frac{\langle n | \nabla \hat{H} | m \rangle \times \langle m | \nabla \hat{H} | n \rangle}{(\epsilon_n - \epsilon_m)^2}$$

例: 自旋在磁场中的运动

$$\hat{H} = -\frac{2\mu_B}{\hbar} \vec{B} \cdot \vec{S}$$

本征值 $E = -2\mu_B B n \quad n = -S, -S+1, \dots, S$

$\vec{B}$ 大小不变, 沿闭合回路 C 缓慢变化一周, 选取参数 θ, φ

$$V_n = \frac{1}{\hbar B} \text{Im} \sum_{m \neq n} \frac{\langle n | \vec{S} | m \rangle \times \langle m | \vec{S} | n \rangle}{(m-n)^2}$$

选取 $\vec{z} \parallel \vec{B}$

$$\begin{aligned} \langle n | \vec{S} | m \rangle \times \langle m | \vec{S} | n \rangle &= \langle n | S_x \vec{e}_x + S_y \vec{e}_y | m \rangle \times \langle m | S_x \vec{e}_x + S_y \vec{e}_y | n \rangle \\ &= (\langle n | S_x | m \rangle \langle m | S_y | n \rangle - \langle n | S_y | m \rangle \langle m | S_x | n \rangle) \vec{e}_z \end{aligned}$$

$$\sum_{m \neq n} \frac{\langle n | \vec{S} | m \rangle \times \langle m | \vec{S} | n \rangle}{(m-n)^2}$$

$$\begin{aligned} &= \sum_{m \neq n} \frac{1}{(m-n)^2} \left[\langle n | \frac{1}{2} (\hat{S}_+ + \hat{S}_-) | m \rangle \langle m | \frac{1}{2i} (\hat{S}_+ - \hat{S}_-) | n \rangle \right. \\ &\quad \left. - \langle n | \frac{1}{2i} (\hat{S}_+ - \hat{S}_-) | m \rangle \langle m | \frac{1}{2} (\hat{S}_+ + \hat{S}_-) | n \rangle \right] \end{aligned}$$

$$\vec{V}_n = n \frac{\vec{B}}{B} \quad \text{磁单极子场}$$

$$\delta_m(C) = - \iint_C \vec{V}_n \cdot d\vec{S}$$

$$= - \iint_C n \frac{B}{B^3} B^2 \sin \theta d\theta d\varphi = -n \Omega(C)$$

拓扑性质 (拓扑不变量)